

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Sixth Semester (Biotechnology) (Credit system) Degree Examinations
May - June 2023

20BTC110 - NANOBIO TECHNOLOGY

Duration: 3 hours

Max.Marks:100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Compare and contrast Bottom up and top down approach of synthesis of nanoparticles with suitable example.	10	L*2	1	2
	b) Discuss on the types of Nanolithography process.	10	L2	1	3
2.	a) Does structure of nanoparticles decide the property of it? Justify with suitable examples.	10	L2	1	2
	b) Compare and contrast Scanning tunnelling microscopy and atomic force microscopy.	10	L2	2	1
3.	a) State the comparison and difference between 0D, 1D and 2D nanomaterials with suitable examples.	10	L2	2	2
	b) Describe the physical, chemical and structural characterization methods of nanomaterial.	10	L2	2	2
Unit – II					
4.	a) Discuss on the clinical considerations for nanotechnology in drug delivery.	10	L2	3	2
	b) Justify the importance of nanotechnology for the delivery of small molecules.	10	L2	3	2
5.	a) State the challenges of Nanopharmaceuticals in clearing the regulatory compliance.	10	L2	3	1
	b) Comment on the different fabrication techniques used for the application of microfluidic channel in diagnosis.	10	L2	4	4
6.	a) Illustrate the preparation and applications of magnetic particles in various fields.	10	L2	4	1
	b) Give the applications microchip based Single cell functional Proteomics.	10	L2	4	1
Unit – III					
7.	a) "Bioconjugated Nanoparticles in Bioanalysis." Justify the statement.	10	L2	5	1
	b) With suitable examples explain the biosignal transduction mechanism.	10	L2	5	1
8.	a) What are transducers? Give the classification of transducers based on function.	10	L2	5	2
	b) "Bio MEMS as drug delivery system." Justify the statement.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
